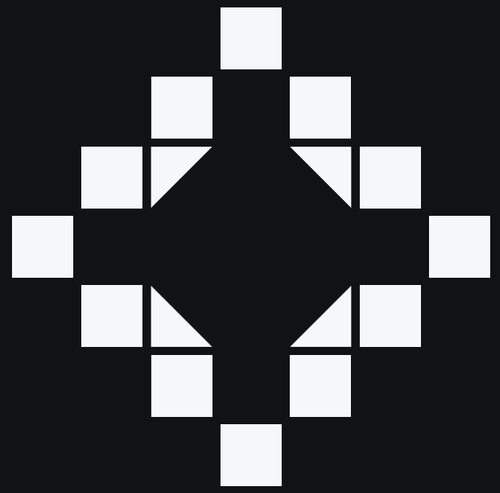
GREENHOUSE ROBOTICS

KRISHAANTH RAMARAJ · FOUNDER

VINEA

Vinea turns the way greenhouse crews work into robot skills, and runs them on one platform inside the greenhouses growers already have.



SOURCE: WAGENINGEN UR, 2005-2010

STILL DONE BY HAND

Greenhouses still run on hands. A hectare of greenhouse tomatoes takes about 9,450 hours of work a year. Of the six main crop jobs, harvesting takes 29% of the hours, removing old leaves 26%, and de-shooting and twisting the plants another 21%. Scouting for pests comes on top. Crews do all of it, all season, and every one of those hours is paid.



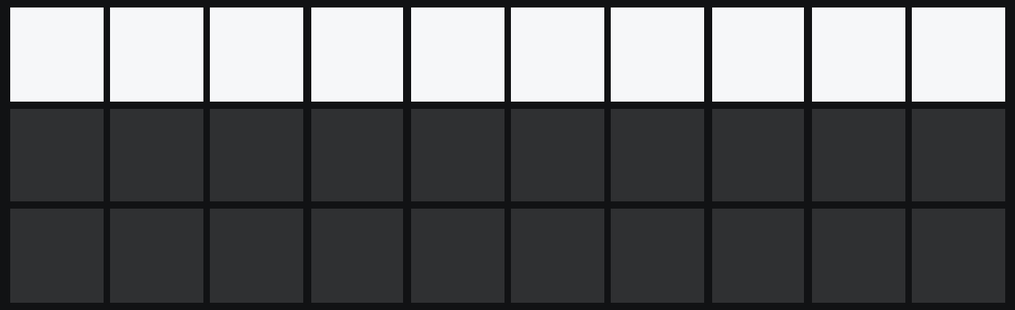
Harvesting BLUSH	29%
Removing old leaves SHEAR	26%
De-shooting and twisting	21%
Clipping, lowering, truss pruning	24%

HOURS ACROSS THE SIX MAIN CROP JOBS, TOMATO

SOURCE: AGRIMATIE (WUR), 2021

A THIRD OF THE BILL

In 2021 the average Dutch greenhouse vegetable farm paid almost €975,000 in wages, out of almost €2.9 million in total costs. One Westland tomato grower told me that harvesting alone costs €5 per m² every year. Since then, costs have risen faster than what growers earn.



WAGES AGAINST TOTAL COSTS, AVERAGE DUTCH GREENHOUSE VEGETABLE FARM, 2021

SOURCE: AGRIMATIE (WUR), 2025

ONE IN FIVE AT A LOSS

In 2025, 20% of Dutch greenhouse vegetable farms made a loss, the first time in more than ten years. Across Dutch greenhouse horticulture, revenue rose 8% while costs and depreciation rose 13%. A grower can't fix that by hiring fewer people, because the crop still needs the same hours.

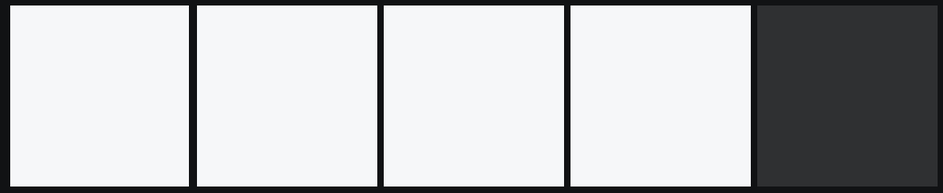


FARMS WITH NEGATIVE INCOME, 2025

SOURCE: RABORESEARCH, 2025

FOUR IN FIVE FROM ABROAD

About 67,000 people work in Dutch greenhouses, and an estimated 80% of them come from abroad, increasingly from Romania and Bulgaria rather than Poland. The area one worker looks after has barely changed since 2000, so a bigger greenhouse still means more people, hired from the same agencies every grower is competing for.



GREENHOUSE WORKERS FROM ABROAD, ESTIMATED

SOURCES: PRESS RELEASES

YEARS IN. MILLIONS IN.

Robots were meant to fix this, and the money has been there for years.

COMPANY	SINCE	RAISED
MetoMotion (GRoW)	2016	A $5M round led by Ridder in 2021
SAIA Agrobotics	2017	Over €20M. The greenhouse is rebuilt around a fixed robot
Four Growers	2018	$15M, with Y Combinator among the backers
eternal.ag	2025	€8M in 2026
Polybot	2025	Backed by SPRIND. Learns truss harvesting from human demonstrations

SOURCES: AGRIDIRECT, WUR, TRADE PRESS

STILL IN TRIALS

In a 2019 survey of over 1,400 Dutch growers, 11% of glasshouse growers used a robot of any kind, mostly for spraying, and 2.6% planned to invest in robotics. In a 2025 WUR test, GRoW picked half the trusses staff marked ripe; 40% of ripe trusses failed its picking conditions, and in 1.5 hours it stopped twice and needed a person six times.

eternal.ag has run one robot at Van Noord Growers since September 2025 and planned three for summer 2026. Four Growers reports up to 50% labour savings at Westburg. Each robot does one job, and the crop has to fit the machine.

THE INSIGHT

SKILL LIVES IN THE CREW

What these robots get wrong are judgement calls: which truss is ready, which leaf comes off, where the whitefly starts. Crews make those calls all day, from experience built up on the job. That judgement lives in the crew, and the crew is the part growers can no longer hire. When a crew leaves at the end of the season, its know-how leaves with it.

SOURCE: AI BUSINESS, NOV 2024

LEARNING FROM PEOPLE

That judgement can now be taught. Robot foundation models learn a new task from recorded demonstrations, and Physical Intelligence alone raised $400 million in 2024 to build them. What these models lack is data from inside working greenhouses, and nobody records it there, because the people with the skill are out on the crop rows.

HOW IT WORKS

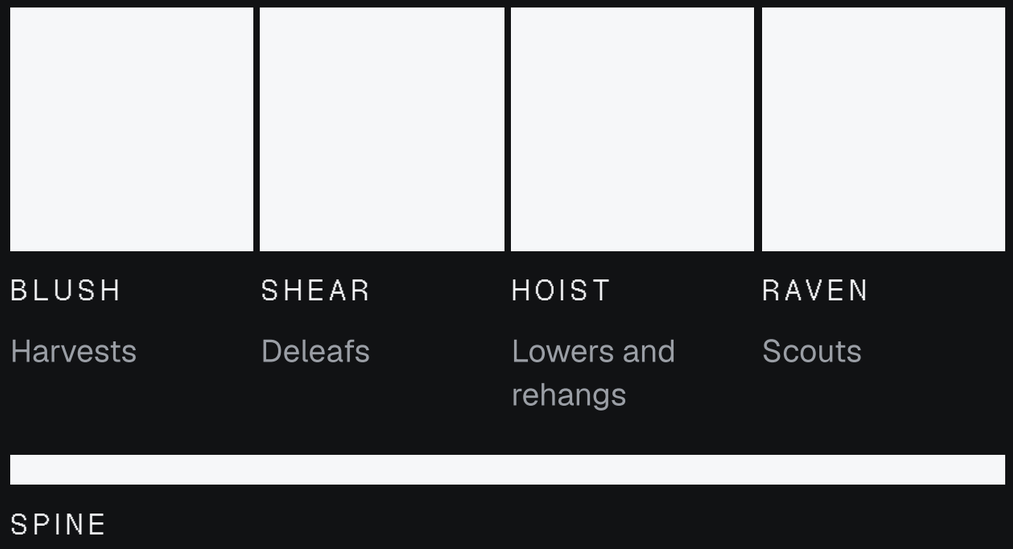
CREWS TEACH THE ROBOTS

Vinea records crews on their normal shifts, and the work itself doesn't change. For harvesting and deleafing, the crew uses a handheld tool shaped like the robot's gripper, with a camera looking along it, so every cut is recorded from the robot's point of view.

For scouting, eye-tracking glasses and a microphone record where an experienced scout looks and what they say they see, so the recording labels itself. Each job's recordings train a robot skill on top of an open robot foundation model.

MODULES FOR EVERY JOB

Every skill runs on SPINE, a base that rides the heating pipes already laid between the crop rows. On top sits the module for the job: BLUSH harvests, SHEAR deleafs, HOIST lowers and rehangs, and RAVEN scouts. A new job means recording the crew doing it and shipping a module. The greenhouse stays as it is, and the grower deals with one platform for the whole crew's work.



THE PRODUCT

HOW BLUSH HARVESTS

BLUSH is the first module, and it harvests whole trusses. A camera facing the row finds a truss that's ready by the grower's own rule: the fruit at the tip has started to colour and most of the rest are ripe. The harvesting model, trained on crew recordings, steers the arm to the stalk through the crowded crop. One close of the fingers cuts the stalk flush with a blade at the tips, and pads keep holding the truss on its way to the crate.

- Find
- Steer
- Cut
- Hold

PROGRESS

FROM SIM TO REAL

On the bench, a robot arm runs a camera model that tells ripe from unripe tomatoes live, on real fruit. In simulation, the arm cuts old leaves in a greenhouse row, harvest training data from an earlier single-tomato version is published openly, and harvesting is moving to whole trusses. Next: train those skills, run them on real plants, and build the capture tool crews will use. Today the demonstrations come from simulation; at scale, crews record them as they work.

IT LEARNS AS IT WORKS

Once deployed, every robot records its own work: each pick, each miss and each time a person steps in, so the whole fleet keeps improving. Each new crop starts from what earlier crops taught it, because picking a pepper has a lot in common with picking a tomato. The more greenhouses Vinea runs in, the less new data each new crop and each new greenhouse needs.

SOURCES: CBS, RABORESEARCH, 2025

DUTCH GLASS FIRST

Vinea starts where the crop, the pipe rails and the crews are the same, and widen from there. Dutch tomatoes cover 1,767 hectares, over half of it farmed by the ten largest growers. Dutch peppers and cucumbers add another 2,213 hectares on the same pipe rails.

Europe has 33,637 hectares of high-tech greenhouses, about half of the world's 70,000. Dutch builders put up glasshouses everywhere, and SPINE follows the ones with pipe rails.

- Dutch tomatoes
- Dutch peppers and cucumbers
- Europe
- The world

SOURCES: RABORESEARCH, CBS

WORLD VS NETHERLANDS

Harvesting alone costs a Westland grower €5 per m², or €50,000 per hectare a year. The world has 3.72 million hectares of greenhouses of every kind. Deleafing, lowering and scouting come on top: in tomatoes, harvesting is under a third of the hours.

MARKET	AREA	HARVESTING, PER YEAR
TAM · World, high-tech glass	About 70,000 ha	€3.5 billion at Dutch cost
SAM · Dutch tomatoes, peppers, cucumbers	3,980 ha	€199 million
SOM · The ten largest Dutch tomato growers	More than 880 ha	Over €44 million

BUSINESS MODEL

RENT IT LIKE A CREW

Growers rent Vinea the way they would hire a crew: a pilot season, then a monthly fee per robot with service included, always priced below the labour it replaces. Pricing stays provisional until after the pilot.

Work out what it saves you →

WHY ME

KRISHAANTH RAMARAJ

At Degla AI, a YC-backed drone startup, I built and tuned drones, precision work of the kind Vinea needs. I made the harvest simulation and I'm building the real robot myself, while studying Electrical Engineering at TU/e and bartending on the side. I've interviewed tomato growers on what labour costs them and why they haven't bought a robot yet, and built Vinea around their answers.

LINKEDIN

LET'S TALK

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THANK YOU

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